Niccolò **Laurenti**

Ph.D. Graduate in Particle Physics · Software Developer

(+39) 3382971956 | niclaurenti@gmail.com | https://niclaurenti.github.io | niclaurenti | niccolo-laurenti |

0009-0001-0718-0409

Summary

Ph.D. in theoretical particle physics, with 3 years of experience in the research field and 1 in industry, specialised in applying artificial intelligence to investigate proton structure, currently working as embedded software developer in the defence sector. I have working experience with different programming languages, in particular with `C++`, `Python`, `Java` and `Ada` and with various machine learning tools like `Keras` and `Tensorflow`.

Personal Informations

Birth	1997, Rome, Italy
Citizenship	Italian
Languages	Italian (native language), English (fluent)

Experience

Embedded Software Developer

Rome, Italy

Embedded Software Developer at Next Ingegneria dei Sistemi S.p.A. — *Oct. 2024 - Current*

- Worked for **Next Ingegneria dei Sistemi** as a consultant software developer for **MBDA** Italy, contributing to a complex defense-related embedded software project.
- Translated high-level system requirements into detailed low-level software specifications, developed software components written mainly in `C++`, `Java` and `Ada` and then designed, performed and automatized corresponding test procedures.
- Participated in system integration activities, interfacing and validating multiple software components across subsystems.
- Collaborated within a large, multi-company team, ensuring cross-organizational alignment and timely delivery of project milestones.

Technologies: C++, Java, Ada, Qt Creator, Apache NetBeans, Gnat Studio, SQL, Bash, XML, IBM RTC, IMB Doors, Wireshark, Linux, Windows, Microsoft Office

Ph.D. Researcher

Milan, Italy

Researcher in Theoretical Particle Physics at the University of Milan and INFN — *Oct. 2021 - Sept. 2024*

- Worked under the supervision of Prof. Stefano Forte in the **NNPDF** collaboration as a developer of the `NNPDF` code.
- Developed techniques and computational programs that utilize artificial intelligence to investigate the internal structure of the proton analysing experimental data collected at **CERN**.
- Developed programs for solving the so-called DGLAP equations, a linear system of integro-differential equations, with numerical techniques.
- Published research results in various papers and presented them at international conferences.

Technologies: Python, Numpy, Scipy, Matplotlib, Keras, Tensorflow, Numba, Fortran, Bash, Git, Github, YAML, JSON, Slurm, PBS, Wolfram Mathematica, Linux, MacOS, VS Code, Vim, Latex, SQL, SQLite

Undergraduate Researcher

Rome, Italy

Researcher in Theoretical Particle Physics at the University of Rome "La Sapienza" — *Mar. 2021 - Oct. 2021*

- Worked under the supervision of Dr. Marco Bonvini to develop theoretical methods and computational programs for producing high-precision theoretical predictions in particle physics.
- Focused on describing experimental data of electron-proton collisions, collected at the particle accelerators **HERA** and **SLAC**.
- Wrote from zero the `C++` library `Adani`, with the `Python` bindings available in the PyPI and in conda-forge, resulting in a published paper and presentations at international conferences.

Technologies: C++, GSL, Wolfram Mathematica, Linux, Bash, CMake, Emacs, Latex

Skills

Programming	C, C++, Python, Java, Ada, Fortran, Bash, XML, YAML, JSON, CMake, SQL
Operating systems	Linux, MacOS, Windows
Code editors	VS Code, Qt Creator, Apache NetBeans, Gnat Studio, Emacs, Vim, Nano
Version control sysytems	Git, Github, Gitlab, Bitbucket, IBM RTC
Python packages	Numpy, Scipy, Matplotlib, Multiprocessing, Numba, Pandas, Keras, Tensorflow
Jobs schedulers	Slurm, PBS
Scientific programs	Matlab, Wolfram Mathematica
Writing	Latex, Markdown, Microsoft Office

Education

Ph.D. in Physics
Milan, Italy

University of Milan — *Oct. 2021 - Nov. 2024*

- Field of study: Theoretical Particle Physics, Computational Physics.
- Thesis: *Advancements in PDFs determination: Incorporation of QED effects and new theoretical improvements in a modern deep learning fitting framework*. **link**

M.S. in Physics
Rome, Italy

University of Rome "La Sapienza" — *Sep. 2019 - Oct. 2021*

- Field of study: Theoretical Particle Physics.
- Grade: 110/110 (cum laude).
- Thesis: *Construction of a next-to-next-to-next-to-leading order approximation for heavy flavour production in deep inelastic scattering with quark masses*. **Inspire**

B.S. in Physics
Rome, Italy

University of Rome "La Sapienza" — *Sep. 2016 - Nov. 2019*

- Grade: 110/110 (cum laude).
- Thesis: *Particle identification with the time of flight method and applications to the CMS experiment*.

Publications

2024	**LO, NLO, and NNLO Parton Distributions for LHC Event Generators**, J. Cruz-Martinez, S. Forte, N. Laurenti, T. R. Rabemananjara, J. Rojo, *JHEP*	***Inspire***
2024	**NNPDF4.0 aN^{3}LO PDFs with QED corrections**, A. Barontini, N. Laurenti, J. Rojo, *Contribution to DIS2024*	***Inspire***
2024	**The Path to N^{3}LO Parton Distributions**, The NNPDF Collaboration, R. D. Ball et al., *Eur. Phys. J. C*	***Inspire***
2024	**Determination of the theory uncertainties from missing higher orders on NNLO parton distributions with percent accuracy**, The NNPDF Collaboration, R. D. Ball et al., *Eur. Phys. J. C*	***Inspire***
2024	**Photons in the proton: implications for the LHC**, The NNPDF Collaboration, R. D. Ball et al., *Eur. Phys. J. C*	***Inspire***
2023	**Inclusion of QED corrections in PDFs fits**, N. Laurenti, *Nucl. Part. Phys. Proc.*	***Inspire***
2022	**Approximating missing higher-orders in transverse momentum distributions using resummations**, N. Laurenti, T. R. Rabemananjara, and R. Stegeman, *Contribution to DIS2022*	***Inspire***

Teaching activity

2024	**Co-supervisor of a Bachelor thesis**, Thesis title: *On the fitting scale dependence of the Parton Distributions*	*University of Milan*
2024	**TA for the course of Quantum Physics I**, Introduction to Quantum Mechanics	*University of Milan*
2024	**TA for the course of Physics**, Basics of Classical Mechanics and Thermodynamics	*University of Milan*
2024	**TA for the course of Quantum Physics II**, Advanced course on Quantum Mechanics	*University of Milan*
2023	**TA for the course of Theoretical Physics I**, Introduction to Quantum Field Theory	*University of Milan*
2023	**TA for the course of Physics**, Basics of Classical Mechanics and Thermodynamics	*University of Milan*
2023	**TA for the course of Quantum Physics II**, Advanced course on Quantum Mechanics	*University of Milan*
2023	**Exercise classes for the course of Quantum Physics II**, Advanced course on Quantum Mechanics	*University of Milan*
2022	**TA for the course of Quantum Physics I**, Introduction to Quantum Mechanics	*University of Milan*
2021	**Student collaboration scholarship**, TA for the Laboratory courses of the first three years	*University of Rome "La Sapienza"*